

Lab04-0
集成CPU核
-IP核设计CPU/IP2CPU

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Course Outline

- 一、实验目的
- 二、实验环境
- 三、实验目标及任务

实验目的

1. 复习寄存器传输控制技术
2. 掌握CPU的核心组成：数据通路与控制单元
3. 设计数据通路的功能部件
4. 进一步了解计算机系统的基本结构
5. 熟练掌握**IP**核的使用方法

实验环境

□ 实验设备

1. 计算机（Intel Core i5以上，4GB内存以上）系统
2. Sword2.0/Sword4.0开发板
3. Xilinx VIVADO2017.4及以上开发工具

□ 材料

无

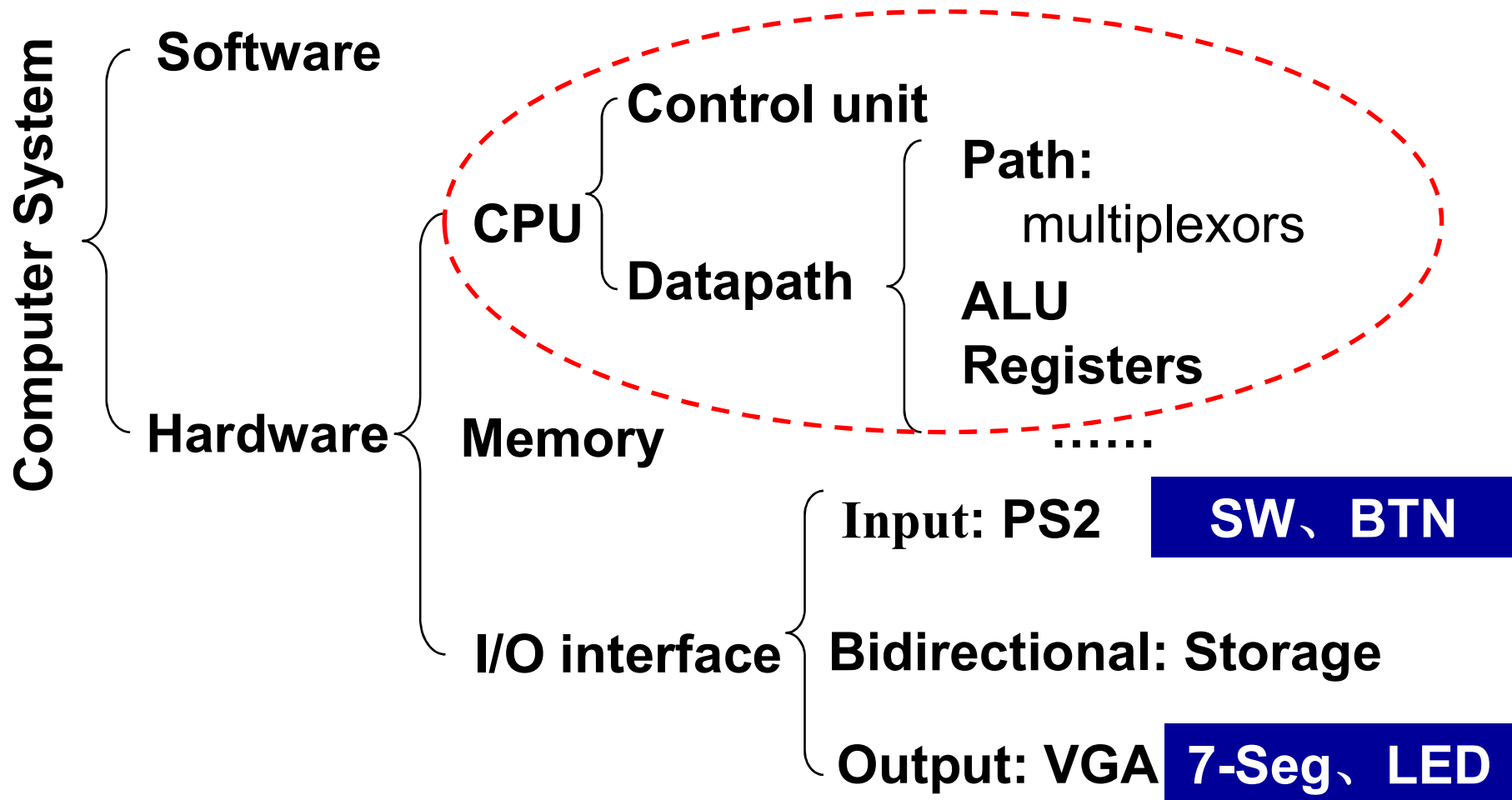
实验目标及任务

- **目标**：熟悉SOC系统的原理，掌握IP核集成设计CPU的方法
- **任务**：利用数据通路和控制器两个IP核集成设计CPU（根据原理图采用RTL代码方式）

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- 任务： 利用数据通路和控制器两个IP核集成设计CPU

Computer Organization

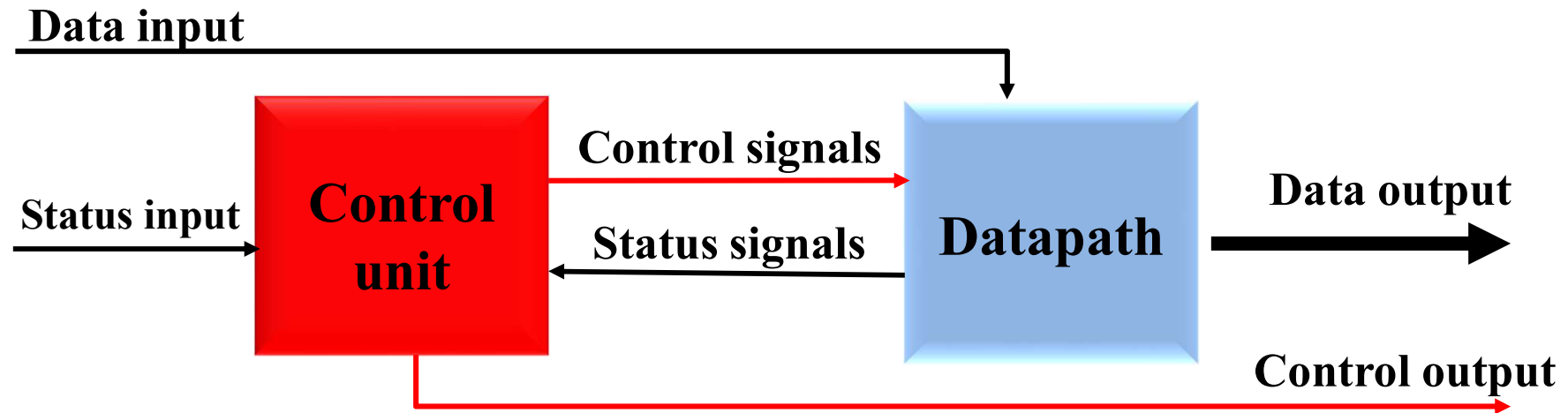
□ Decomposability of computer systems



Digital circuits vs CPU organization

□ Digital circuit

- General circuits that controls logical event with logical gates -
-Hardware



□ Computer organization

- Special circuits that processes logical action with instructions
-Software

CPU部件之1-数据通路: **Data_path**

□ **Data_path**

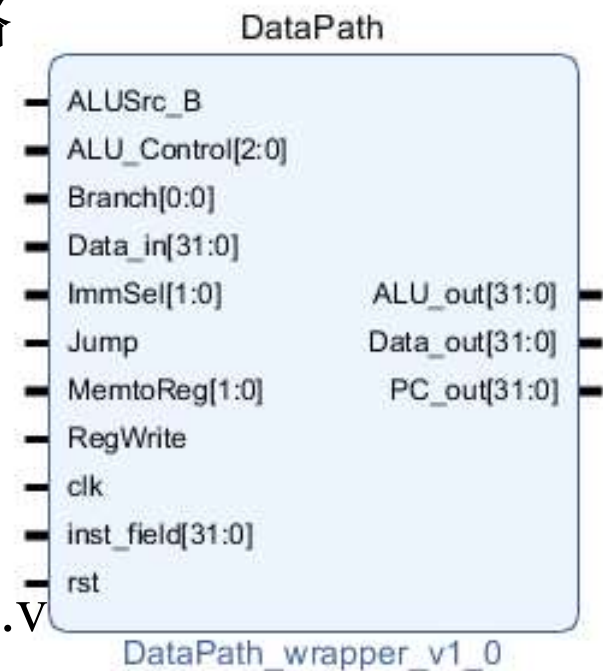
- CPU主要部件之一
- 寄存器传输控制对象: 通用数据通路

□ 基本功能

- 具有通用计算功能的算术逻辑部件
- 具有通用目的寄存器
- 具有通用计数所需的尽可能多的路径

□ 本实验用IP 软核- **Data_path**

- 核调用模块Data_path.v
- 核接口信号模块(空文档): Data_path.v



数据通路空模块- **Data_path.v**

```
module      Data_path( input clk,           //时钟
                        input rst,          //复位
                        input[31:0]inst_field, //指令数据域[31:7]
                        input ALUSrc_B,
                        input [1:0]MemtoReg,
                        input [1:0] ImmSel,
                        input Jump,
                        input Branch,
                        input RegWrite,
                        input[31:0]Data_in,
                        input[2:0]ALU_Control,

                        output[31:0]ALU_out,
                        output[31:0]Data_out,
                        output[31:0]PC_out
                        );

endmodule
```

CPU部件之2-控制器： **SCPU_ctrl**

□ SCPU_ctrl

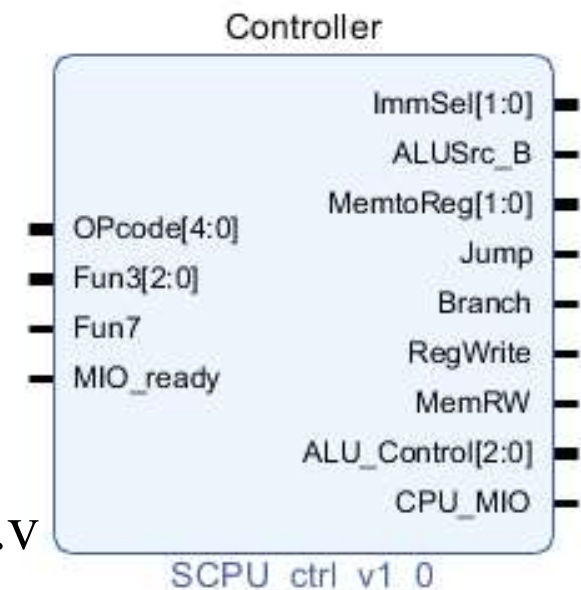
- CPU主要部件之一
- 寄存器传输控制技术中的运算和通路控制器：

□ 基本功能

- 指令译码
- 产生操作控制信号：ALU运算控制
- 产生指令所需的路径选择

□ 本实验用IP 软核- **SCPU_ctrl**

- 核调用模块SCPU_ctrl.v
- 核接口信号模块(空文档)： SCPU_ctrl.v



控制器接口文档- **SCPU_ctrl.v**

```
module      SCPU_ctrl( input[4:0]OPcode,    //Opcode—inst[6:2]
                        input[2:0]Fun3,      //Function-inst[14:12]
                        input    Fun7,      //Function-inst[30]
                        input MIO_ready,    //CPU Wait
                        output reg [1:0] ImmSel      //立即数选择
                        output reg ALUSrc_B,
                        output reg[1:0]MemtoReg,
                        output reg Jump,
                        output reg Branch,
                        output reg RegWrite,
                        output reg MemRW,
                        output reg [2:0]ALU_Control,
                        output reg CPU_MIO
                        );

endmodule
```

RTL集成设计实现CPU

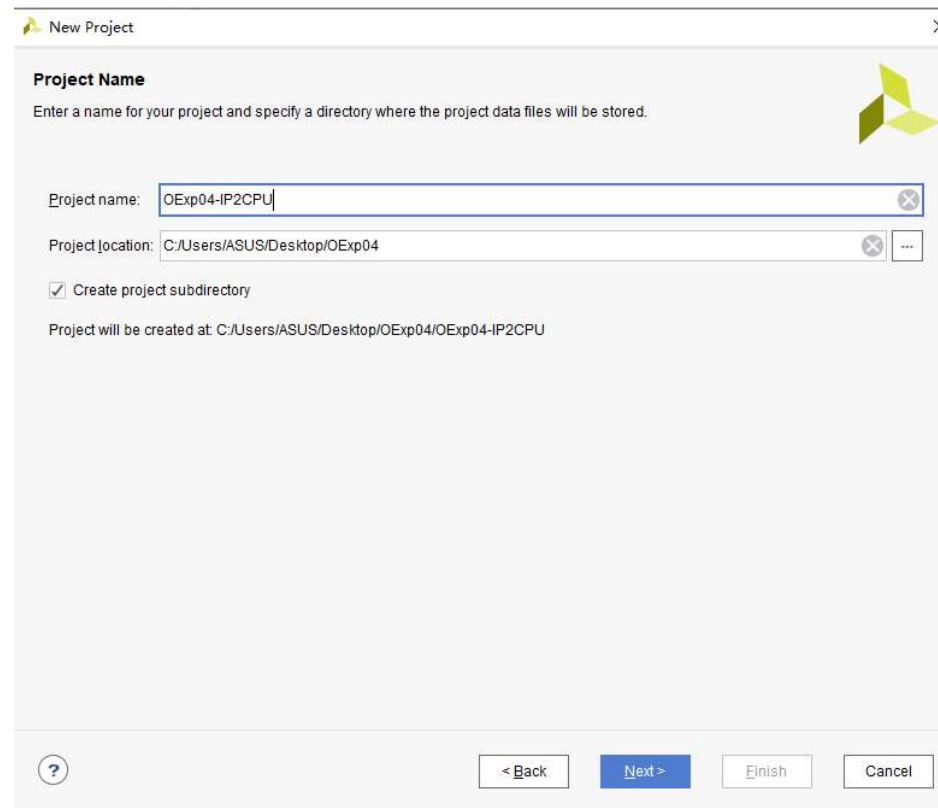
设计工程：OExp04-IP2CPU

◎ 分解CPU为二个IP核

⌘ 用二个IP核构建CPU

⌘ 顶层工程名称沿用OExp04-IP2CPU

◎ 顶层模块名：SCPU.v



The image shows a 'New Project' dialog box from a software development environment. The dialog has a title bar with a yellow icon and a close button. The main area is titled 'Project Name' and contains the instruction 'Enter a name for your project and specify a directory where the project data files will be stored.' Below this, there are two input fields: 'Project name:' with the text 'OExp04-IP2CPU' and 'Project location:' with the text 'C:/Users/ASUS/Desktop/OExp04'. To the right of the location field is a button with three dots. Below these fields is a checkbox labeled 'Create project subdirectory' which is checked. At the bottom, it says 'Project will be created at: C:/Users/ASUS/Desktop/OExp04/OExp04-IP2CPU'. The bottom of the dialog has a row of buttons: a help button (question mark in a circle), '< Back', 'Next >' (highlighted in blue), 'Finish', and 'Cancel'.

New Project

Project Name
Enter a name for your project and specify a directory where the project data files will be stored.

Project name: OExp04-IP2CPU

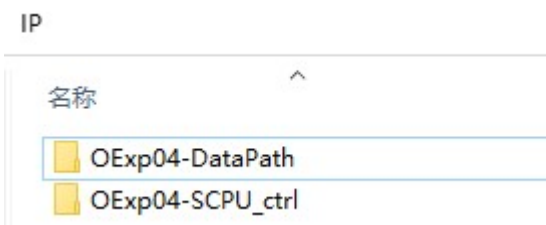
Project location: C:/Users/ASUS/Desktop/OExp04

☒ Create project subdirectory

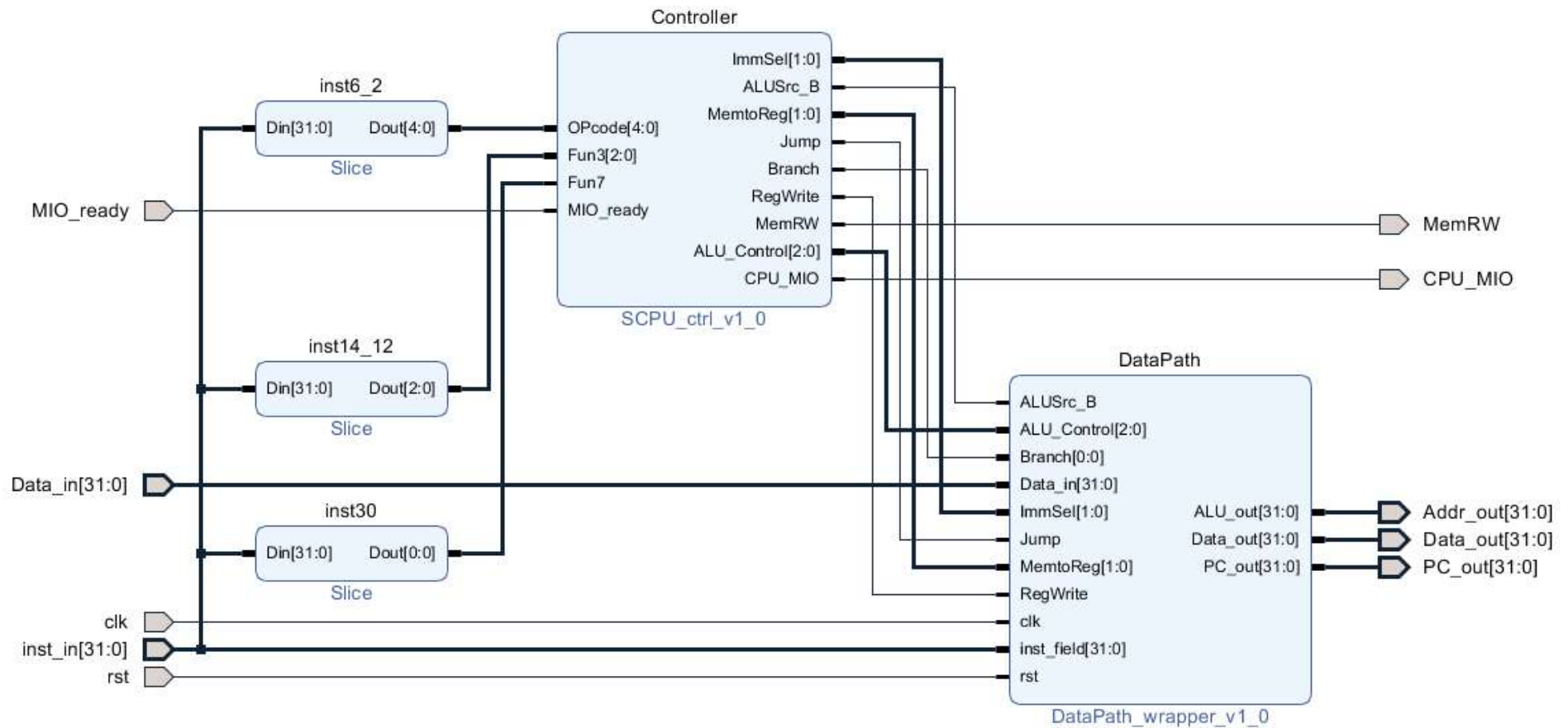
Project will be created at: C:/Users/ASUS/Desktop/OExp04/OExp04-IP2CPU

? < Back Next > Finish Cancel

添加二个核的封装文件到当前工程IP库目录：



逻辑原理图



用RTL代码描述CPU设计 SCPU.V

- 利用verilog描述，在SCPU.v顶层进行模块调用和连接

```
.....  
.....  
module SCPU(  
    input wire clk,  
    input wire rst,  
    input wire MIO_ready,  
    input wire[31:0] inst_in,  
    input wire[31:0] Data_in,  
  
    output wire CPU_MIO,  
    output wire MemRW,  
    output wire[31:0] PC_out,  
    output wire[31:0] Data_out,  
    output wire[31:0] Addr_out  
);  
  
    .....  
    .....  
  
    SCPU_ctrl1 U1(  
        .OPcode      (inst_in[6:2] ),  
        .Fun3         (inst_in[14:12]),  
        .Fun7         (inst_in[30]  ),  
        .MIO_ready    (MIO_ready   ),  
        .ImmSel       (ImmSel       ),  
        .ALUSrc_B     (ALUSrc_B     ),  
        .MemtoReg     (MemtoReg     ),  
        .Jump         (Jump         ),  
        .Branch       (Branch       ),  
        .RegWrite     (RegWrite     ),  
        .MemRW        (MemRW        ),  
        .ALU_Control1 (ALU_Control1 ),  
        .CPU_MIO      (CPU_MIO      )  
    )  
    .....  
    .....
```

CPU集成设计后的层次关系

Exp04完成CPU设计后的
模块调用关系

采用直接添加
源文件的方式

```
40 wire Branch;
41 wire RegWrite;
42 wire [2:0] ALU_Control;
43
44 SCPU_ctrl U1(
45     .OPcode      (inst_in[6:2] ),
46     .Fun3         (inst_in[14:12]),
47     .Fun7         (inst_in[30] ),
48     .MIO_ready    (MIO_ready ),
49     .ImmSel       (ImmSel ),
50     .ALUSrc_B     (ALUSrc_B ),
51     .MemtoReg     (MemtoReg ),
```

◎ **END**